

The gravitational field equations

13

The key idea of general relativity is that the curvature of spacetime, which gives rise to gravitation, is determined by the energy density of the matter fields of the Universe, such that we can write the Einstein equation

$$\left(\begin{array}{c} \text{Curvature of} \\ \text{spacetime} \end{array} \right) = \left(\begin{array}{c} \text{Energy density} \\ \text{of matter fields} \end{array} \right). \quad (13.1)$$

At last, in this chapter, we are able to formulate the details of this equation! Our task is, on the (geometrical) left-hand side, to identify a suitable expression for the curvature of spacetime and, on the (physical) right, a suitable expression for the density of energy in matter fields.

In order to get to Einstein's equation, we take the route shown in Fig. 13.1. Starting with the physical principles of fields and of eqn 13.1, we shall review the key elements of geometry that supply the left-hand side of the equation. We then examine the physics of the right-hand side, in particular the energy-momentum tensor field $\mathbf{T}(x)$ that encodes the energy density of the matter fields that fill the Universe. Finally we link these to form the **Einstein field equation**, the tensor field version of eqn 13.1.

13.1 Geometry: a recap of the key ingredients

The metric tensor field $\mathbf{g}(x)$ is the foundation of the curvature of spacetime. As we've written it, it is a classical field: a function of position in spacetime x that gives us a tensor valid at the point x . We often express the components of the metric field $g_{\mu\nu}(x)$ via a line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (13.2)$$

There is no way to tell whether a single particle accelerates due to the effects of a homogeneous gravitational field or because of the choice of coordinates. If a gravitational field is present, and all real-life gravitational fields are necessarily inhomogeneous, spacetime is curved and this gives rise to the acceleration of a particle in the absence of forces external to spacetime. However, a particle might accelerate merely due to the perversity of the coordinate system that we've chosen, rather than due to curvature. The method to unambiguously detect curvature

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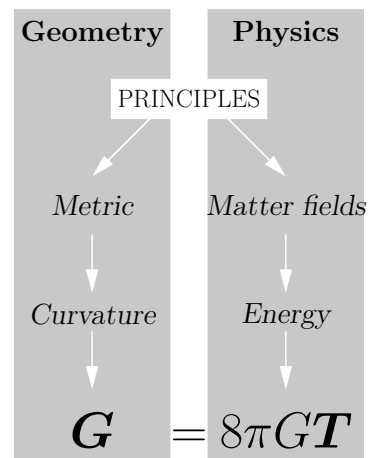


Fig. 13.1 A conceptual route to Einstein's equation. The left-hand side leads to the Einstein tensor field $\mathbf{G}$, related to the Riemann tensor field as described later in this chapter. The right-hand side leads to a constant $8\pi G$ (where G here is the gravitational constant) multiplied by the energy-momentum tensor field $\mathbf{T}$.

¹The connection coefficients may be related to the metric tensor field via the component equation

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2}g^{\mu\lambda} \left(\frac{\partial g_{\lambda\alpha}}{\partial x^\beta} + \frac{\partial g_{\lambda\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\lambda} \right). \quad (13.3)$$

²We use

$$\frac{D^2 \xi}{d\tau^2} = \mathbf{R}(\mathbf{u}, \xi, \mathbf{u}), \quad (13.4)$$

where ξ is the particles' separation and $\mathbf{u}$ is the velocity of the particle following the fiducial geodesic.

³Note for later in the chapter that this is symmetric: $R_{\mu\nu}(x) = R_{\nu\mu}(x)$.

⁴Recall that a scalar is the same, independent of the coordinate system in which it is evaluated.

⁵One point of view is that the (1,3) Riemann curvature tensor field $\mathbf{R}(x)$ is the gravitational field. So although we have the equivalence principle telling us that a measurement at a point cannot tell us whether it's gravitation or acceleration causing an effect, we can identify the physical field that causes gravitation. We cannot simply slot this 'gravitational field' $\mathbf{R}$ into the Einstein equation (curvature) = (energy), owing to the (1,3) valence of $\mathbf{R}$ compare to the (2,0) valence of the energy-momentum tensor $\mathbf{T}$. However, if we are looking for the physical manifestation of gravitation, this is as good a candidate as any.

⁶See also Section 0.4.

involves transforming away the effects due to the coordinate system to arrive at what's left behind: the curvature due to gravitation. This is the motivation behind the Riemann curvature tensor $\mathbf{R}(x)$, which evaluates the presence of this stuff that's left behind. We expect that this tensor should play a key role in the left-hand side of the equation, which indeed it does. It depends on the (first and second) derivatives of the metric field. The first derivatives of $\mathbf{g}(x)$ provide the connection coefficients $\Gamma^\mu_{\alpha\beta}(x)$,¹ and reflect how the metric causes the coordinates to change as we move through spacetime. They are easily manipulated, so we often describe $\mathbf{R}(x)$ in terms of them, rather directly through $\mathbf{g}(x)$.

To obtain $\mathbf{R}(x)$ and thus determine the curvature of spacetime, we evaluate the geodesic deviation of two freely falling particles whose motion was initially parallel and extract $\mathbf{R}(x)$ from the equation of motion of their separation.² The resulting Riemann tensor field has components

$$R^\alpha_{\beta\gamma\delta}(x) = \frac{\partial \Gamma^\alpha_{\delta\beta}(x)}{\partial x^\gamma} - \frac{\partial \Gamma^\alpha_{\gamma\beta}(x)}{\partial x^\delta} + \Gamma^\alpha_{\gamma\mu}(x)\Gamma^\mu_{\delta\beta}(x) - \Gamma^\alpha_{\delta\mu}(x)\Gamma^\mu_{\gamma\beta}(x). \quad (13.5)$$

This gives us access to the curvature at each point in spacetime. Two quantities derived from the Riemann tensor field will be important in setting up the Einstein equation. The first is the Ricci tensor field, which at a spacetime point x has components³

$$R_{\mu\nu}(x) = R^\alpha_{\mu\alpha\nu}(x). \quad (13.6)$$

The second is the Ricci scalar field⁴

$$R(x) = g_{\mu\nu}R^{\mu\nu}(x). \quad (13.7)$$

Using these two questions, we will be able to gain access to the curvature of spacetime. This will be achieved by taking a double derivative of the metric field $\mathbf{g}(x)$, which is the basis of the left-hand side of the Einstein equation.⁵ After having reviewed the geometrical left-hand side of the equation, let's now turn to the right-hand side and the physical content of the theory.

13.2 Physics: the key ingredients

We start with a principle: *a physical relativistic/geometrical field theory of gravitation must be compatible with Newton's theory in the non-relativistic limit*. It is therefore useful to revisit Newton's law of gravitation from the point of view of fields and see what it teaches us about the shape that our new theory must take.⁶ Newton's law says that the force $\vec{F}$ experienced by a mass m from a gravitating object with mass M separated by a displacement vector $\vec{r}$, is given by

$$\vec{F}(r) = -\frac{GMm}{r^3}\vec{r}, \quad (13.8)$$

where $G = 6.672(4) \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$ is the gravitational constant. We also find it useful to deal with a scalar potential energy function

$U(r)$ experienced by mass m due to mass M and given by

$$U(r) = -\frac{GMm}{r}, \quad (13.9)$$

where $\vec{F} = -\vec{\nabla}U$. The function $U(r)$ is proportional to m , so it is better to think about the potential $\Phi(r) = U(r)/m$, the potential energy per unit mass. Newton's law of gravitation can therefore be turned into a field equation straightforwardly. The field version of the law describes the force per unit mass $\vec{g}(\vec{x})$ experienced at a position $\vec{x}$ due to the presence of a distribution of mass with density $\rho(\vec{x}')$. The generalization of the force law is

$$\vec{g}(\vec{x}) = -G \int d^3x' \rho(\vec{x}') \frac{(\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3}. \quad (13.10)$$

We also define a gravitational potential

$$\Phi(\vec{x}) = -G \int d^3x' \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|}. \quad (13.11)$$

We can then show that (i) the function Φ obeys the important rule $\vec{g}(\vec{x}) = -\vec{\nabla}\Phi(\vec{x})$, and, most importantly, (ii) Φ can be computed from a given distribution of mass. This is the purpose of the next example.

Example 13.1

(i) Using the potential $\Phi(\vec{x})$ we can write⁷

$$\vec{g}(\vec{x}) = \vec{\nabla}_{\vec{x}} \int d^3x' \frac{G\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} = -\vec{\nabla}\Phi(\vec{x}). \quad (13.13)$$

(ii) Next, we use the result⁸

$$\vec{\nabla}_{\vec{x}} \cdot \left(\frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} \right) = 4\pi\delta^{(3)}(\vec{x} - \vec{x}'), \quad (13.15)$$

which allows us to take the divergence of $\vec{g}$ in eqn 13.10 to find

$$\begin{aligned} \vec{\nabla} \cdot \vec{g}(\vec{x}) &= -4\pi G \int d^3x' \rho(\vec{x}') \delta^{(3)}(\vec{x} - \vec{x}') \\ &= -4\pi G \rho(\vec{x}). \end{aligned} \quad (13.16)$$

Since we also have $\vec{\nabla} \cdot \vec{g}(\vec{x}) = -\vec{\nabla}^2\Phi$, we conclude that

$$\vec{\nabla}^2\Phi(\vec{x}) = 4\pi G\rho(\vec{x}). \quad (13.17)$$

This is the gravitational version of Poisson's equation.⁹

To recap the main results of the last example, we have that, expressed as a field equation, Newton's force law is written as¹⁰

$$-\vec{\nabla} \cdot \vec{g}(\vec{x}) = 4\pi G\rho(\vec{x}), \quad (13.18)$$

where $\vec{g}$ is the gravitational field. The law can be expressed in terms of the potential Φ (via $\vec{g} = -\vec{\nabla}\Phi$). The potential field is the solution of Poisson's equation

⁷Proof: Check by a direct computation that

$$\vec{\nabla}_{\vec{x}} \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -\frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3}. \quad (13.12)$$

⁸See exercises for the proof that

$$\vec{\nabla}_{\vec{x}}^2 \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -4\pi\delta^{(3)}(\vec{x} - \vec{x}'). \quad (13.14)$$

⁹Recall that Laplace's equation for a field in the absence of sources says $\vec{\nabla}^2\Phi = 0$, but Poisson's equation for the field in the presence of sources says $\vec{\nabla}^2\Phi = 4\pi G\rho$, where $4\pi G\rho$ is the density of the source.

¹⁰Note that eqn 13.18 is the same as presented in eqn 0.12, and eqn 13.19 is the same as presented in eqn 0.15.

¹¹The idea of the Green's function is that for a differential equation $\hat{L}A(x) = f(x)$, with $\hat{L}$ representing a differential operator, we define the Green's function as $\hat{L}G(x, y) = \delta(x - y)$. This is useful, as the Green's function can then be used to build a solution using the prescription

$$A(x) = \int dy G(x, y) f(y). \quad (13.20)$$

¹²We know that both the energy density and metric field are tensor fields, so we might expect Newton's law to be equivalent to one component of the Einstein equation.

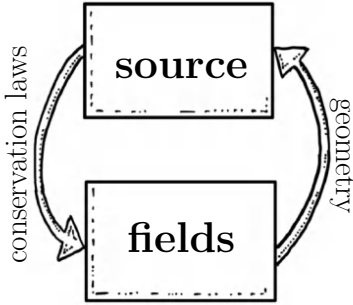


Fig. 13.2 The relationship between the source of energy-momentum $\mathbf{T}$ and the metric field $\mathbf{g}$ in the Einstein equation. The source is constrained by conservation laws (via $\nabla \cdot \mathbf{T} = 0$) affecting its coupling to the metric field that, in turn, determines the geometry. The metric field also determines the form of $\mathbf{T}$ itself.

¹³We can be a little more specific here. In electromagnetism (another linear theory), we often specify the currents and charges and then solve Maxwell's equations to find the electromagnetic fields. The same can be done in Newtonian gravity: specify the mass distribution and solve for Φ . The same cannot be done in general relativity since the source of gravitation is the energy-momentum tensor $\mathbf{T}$, which is described in terms of the metric $\mathbf{g}$, precisely the field we are trying to find. This forces us to attempt to find consistent values of $\mathbf{g}$ and $\mathbf{T}$ simultaneously, which is very difficult.

$$\vec{\nabla}^2 \Phi(\vec{x}) = 4\pi G \rho(\vec{x}). \quad (13.19)$$

Note the form of this equation that says that two derivatives of the potential field are proportional to the mass density ρ , which is the source of the field. We therefore have a field theory of Newtonian gravitation that deals in a scalar field $\Phi(\vec{x}, t)$. We input a position in spacetime and output a function Φ . This can be used to compute the gravitational field and hence the force on any mass.

Example 13.2

Green's functions are a very useful tool in solving equations like the Poisson equation.¹¹ For our problem, we seek a solution to the equation

$$\vec{\nabla}^2 G(\vec{x}, \vec{x}') = 4\pi \delta^{(3)}(\vec{x} - \vec{x}'). \quad (13.21)$$

We can then use the resulting Green's function $G(\vec{x}, \vec{x}')$ to build a solution to Poisson's equation using the prescription

$$\Phi(\vec{x}) = G \int d^3x' G(\vec{x}, \vec{x}') \rho(\vec{x}'). \quad (13.22)$$

Comparing to the steps above, we can identify the Green's function we need to solve Poisson's equation: it is simply

$$G(\vec{x}, \vec{x}') = -\frac{1}{|\vec{x} - \vec{x}'|}. \quad (13.23)$$

We shall try to fashion our theory of gravitation following the pattern set by Newton's field theory. In place of the Newtonian potential, we shall substitute the metric. In place of the mass density, we substitute the energy density. If we postulate that the Einstein equation has the same form as Poisson's equation for the Newtonian potential field, then Einstein's equation must look like

$$\boxed{\partial^2 g} = \boxed{\kappa T}, \quad (13.24)$$

where κ is a constant.¹² That is to say that two derivatives of the metric field g are proportional to the energy density T .

Although these steps might make us optimistic, there must also be differences between Einstein and Newton's pictures of gravitation. First, Newton's law is instantaneous, running counter to the rules of special relativity. In addition, Fermat's principle says that light propagates in straight lines in flat spacetime but experiments show the bending of light from stars by gravitation, implying that gravity must warp spacetime. This means that, for Einstein's gravity, the geometry of curved space is caused by the energy density of matter which acts as a source (just like ρ is a source of Φ). However, the arrangement of energy density is itself determined by the geometry encoded in the metric field, leading to the situation shown in Fig. 13.2. We conclude that, in contrast to Newton, Einstein's gravitation is necessarily nonlinear: gravitational energy is acted on by gravity itself.¹³

We have now reached the point where we must fit the geometrical and physical fields together. The key physical principles determining the content on the right-hand side of a *relativistic* equation of motion are as follows. (i) **Local causality**: A signal can be sent between two points if, and only if, their separation is not spacelike. Colloquially: nothing travels faster than light. Our fields must therefore be consistent with special relativity. (ii) **Local conservation of energy and momentum**: All matter fields in the Universe have energy-momentum. This energy-momentum is locally conserved. That is to say, it must obey a continuity equation.

We also require energy to obey another condition. (iii) The **dominant energy condition** says that to any observer the local energy density appears non-negative and the local energy flow vector is not spacelike (i.e. it is timelike or null). We therefore demand that positive energy, which cannot move around a system at a speed faster than that of light, is locally conserved.¹⁴

All of the physical principles relating to the behaviour of mass energy can be embodied in an **energy-momentum tensor field** $\mathbf{T}(x)$, as described in the previous chapter. This tensor must describe positive energy and encodes local conservation of energy-momentum (at least approximately) via the constraint $\nabla \cdot \mathbf{T} = 0$. The energy-momentum tensor vanishes only if all of the matter fields vanish. This means that all fields cost energy, and this is the source of gravitation.

In the next section, we shall, finally, attempt to marry the geometry of curvature embodied in $\mathbf{R}$ and the mass energy of the fields of the Universe, expressed in $\mathbf{T}$ in a tensor equation.

¹⁴There is a less stringent **weak energy condition**, which says that the energy density measured by any observer must be non-negative.

13.3 An incorrect guess

The apparently obvious field equation turns out to be incorrect.

The energy-momentum tensor has two slots or, equivalently, its components have two indices. We are therefore tempted to guess that the Einstein equation, that fixes the relationship of curvature (on the left) to the energy density of the matter fields (on the right) has a (2,0) object on the left and $T^{\mu\nu}$ on the right. The obvious, but incorrect, natural candidate would be that the curvature-defining object is the Ricci tensor, with components $R^{\mu\nu}$. This would give us the field equation

$$R^{\mu\nu} = \kappa T^{\mu\nu} \quad (\text{incorrect guess}), \quad (13.25)$$

with κ a constant.¹⁵ But why is this wrong? The key is to remember that we must have the conservation equation $\nabla \cdot \mathbf{T} = 0$. We examine the consequence of this on our candidate field equation by taking the divergence

$$R^{\mu\nu}{}_{;\mu} = \kappa T^{\mu\nu}{}_{;\mu} \quad (\text{incorrect guess}). \quad (13.26)$$

Since $\nabla \cdot \mathbf{T} = 0$ (or, equivalently $T^{\mu\nu}{}_{;\mu} = 0$), the right-hand side of eqn 13.26 must vanish. What happens to the geometrical left-hand side?

¹⁵ This seems a good idea since the Ricci tensor is a sort of average of $\mathbf{R}$ that retains those parts of curvature that, in a gravitational equation, can cause volumes to shrink. This was Einstein's original suggestion in October 1915.

To answer this, we need a mathematical result to constrain our equations.

Example 13.3

As we'll discuss in Chapter 43, the components of the Riemann tensor satisfy a geometrical constraint known as the **Bianchi identity**, whose component form is written as

$$R^\alpha{}_{\beta\gamma\delta;\lambda} + R^\alpha{}_{\beta\lambda\gamma;\delta} + R^\alpha{}_{\beta\delta\lambda;\gamma} = 0. \quad (13.27)$$

For now, it is enough to say that this identity comes from a general principle that 'the boundary of a boundary is zero', an idea that leads to the well-known vector identity that the divergence of the curl of a vector field vanishes. We can contract indices in this identity, setting $\alpha = \lambda$, to find¹⁶

$$R^\alpha{}_{\beta\gamma\delta;\alpha} + R^\alpha{}_{\beta\gamma;\delta} - R^\alpha{}_{\beta\delta;\gamma} = 0. \quad (13.28)$$

Now raise the β index and contract with the δ index to get

$$\begin{aligned} R^\alpha{}_{\gamma;\alpha} + R^\beta{}_{\gamma;\beta} - R_{;\gamma} &= 0 \\ 2R^\alpha{}_{\gamma;\alpha} &= R_{;\gamma}. \end{aligned} \quad (13.29)$$

This equation is known as the **contracted Bianchi identity** and we will use it below.

To examine the consequence of the identity discussed in the last example, we now define the **trace-reversed Ricci tensor**, also known as the **Einstein tensor field**, as the (2,0) tensor $\mathbf{G}(x)$ with components¹⁷

$$G^{\mu\nu}(x) = R^{\mu\nu}(x) - \frac{1}{2}g^{\mu\nu}(x)R(x). \quad (13.30)$$

Taking the divergence of the Einstein tensor $\nabla \cdot \mathbf{G}$, we find that¹⁸

$$G^{\mu\nu}{}_{;\mu} = R^{\mu\nu}{}_{;\mu} - \frac{1}{2}g^{\mu\nu}R_{;\mu}. \quad (13.31)$$

Using the contracted Bianchi identity ($R_{;\mu} = 2R^\alpha{}_{\mu;\alpha}$) on the second term on the right, we find¹⁹

$$G^{\mu\nu}{}_{;\mu} = R^{\mu\nu}{}_{;\mu} - g^{\mu\nu}R^\alpha{}_{\mu;\alpha} = 0. \quad (13.32)$$

We conclude that the Einstein tensor has the property

$$\nabla \cdot \mathbf{G} = \left(R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R \right)_{;\mu} = 0. \quad (13.33)$$

Although this doesn't look particularly scandalous, it causes a major problem for the divergence of our candidate equation: $R^{\mu\nu}{}_{;\mu} = \kappa T^{\mu\nu}{}_{;\mu} = 0$. We combine this expression with eqn 13.33 to conclude

$$-\frac{1}{2}g^{\mu\nu}R_{;\mu} = -\frac{1}{2}R^{;\nu} = 0. \quad (13.34)$$

The guts of this latter equation²⁰ ($R^{;\nu} = 0$) forces the Ricci scalar field $R(x)$ to be a constant throughout the Universe. This on its own is not

¹⁶We make use of the symmetries of $R_{\alpha\beta\gamma\delta}$, where swaps within the first and second pairs of indices result in a minus sign.

¹⁷Do not confuse the Einstein tensor field $\mathbf{G}$ with the gravitational constant G . It's unfortunate that they have the same symbol, but context should make clear which is which. In later chapters, we'll use units in which $G = 1$.

¹⁸We use the compatibility condition $g_{\mu\nu;\alpha} = 0$ here.

¹⁹If you don't see this immediately, the steps are (i) raise the μ index in the second term to find $R^{\mu\nu}{}_{;\mu} - R^{\alpha\nu}{}_{;\alpha}$; (ii) rename the α index μ and the expression vanishes.

²⁰Remember that the effect of the covariant derivative on a scalar field is simply

$$\nabla f = f_{;\mu} = f_{,\mu} = \frac{\partial f}{\partial x^\mu}.$$

a problem. However, the trouble comes when we examine the candidate equation (eqn 13.25) again, since we find that

$$R(x) = g^{\mu\nu}(x)R_{\mu\nu}(x) = \kappa g^{\mu\nu}(x)T_{\mu\nu}(x) = \text{const.} \quad (13.35)$$

That is, the trace $g^{\mu\nu}T_{\mu\nu}$ of $T_{\mu\nu}$ must be constant throughout the Universe. *This* is unreasonable! It implies a uniform distribution of matter and energy throughout the Universe which is at odds with our experience, where matter is certainly non-uniform. We are forced to drop the candidate equation.

13.4 Einstein's field equation

The problem encountered in the last section leads to the solution.

On our way to proving that our first attempt at a field equation was inadequate, we identified a new (2,0) [or (0,2)] tensor $\mathbf{G}$, which had components $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$. It is this object that provides the solution to our problem. Therefore, having discarded our first guess, we now consider instead a new candidate field equation²¹

$$G_{\mu\nu} = \kappa T_{\mu\nu}. \quad (13.36)$$

We know from the last section that the divergence of this equation is zero on both sides. This, it turns out, is exactly the field equation for which we've been searching. However, we have yet to work out the constant of proportionality κ . This can be evaluated using the condition that this equation is compatible with the results of Newtonian gravitation in the limit of weak fields (i.e. with the equation $\vec{\nabla}^2\Phi = 4\pi G\rho$).

Example 13.4

In the weak-field limit, spacetime is only very slightly curved and therefore the metric is very close to the flat Minkowski metric, so $g_{00} \approx g^{00} \approx -1$ and $g_{ii} \approx g^{ii} \approx 1$. As a result, we have for any tensor $\mathbf{A}$ in this geometry, that $A^{00} \approx A_{00}$, along with $A^{ii} = A_{ii}$. Moreover, the energy-momentum tensor is dominated by the energy density part (because $\rho \gg p/c^2$) so the only significant element of $\mathbf{T}$ is $T^{00} = \rho c^2$, with all other elements vanishingly small. This means that, for all of the spatial (ij) components of the Einstein tensor, we have

$$G^{ij} = R^{ij} - \frac{1}{2}g^{ij}R = 0, \quad (13.37)$$

from which we find $R^{11} = R^{22} = R^{33} = R/2$ and so

$$R^{11} + R^{22} + R^{33} = 3R/2. \quad (13.38)$$

As a result, $R = g_{\mu\nu}R^{\mu\nu} = -R^{00} + \frac{3}{2}R$ and therefore $R = 2R^{00}$. This allows us to calculate $G^{00} = R^{00} - \frac{1}{2}g^{00}R$ which gives

$$G^{00} = 2R^{00} \approx 2R_{00}. \quad (13.39)$$

Recall that, with the weak-field metric, the only appreciable connection coefficients are (replacing factors of c) $\Gamma_{00}^i = \partial(\Phi/c^2)/\partial x^i$ and so $R^i_{0j0} = \partial^2(\Phi/c^2)/\partial x^i\partial x^j$. We can then read off²² that $R_{00} = (1/c^2)\vec{\nabla}^2\Phi$ and so

$$G^{00} = \frac{2}{c^2}\vec{\nabla}^2\Phi. \quad (13.40)$$

²¹In case this mathematics is a distraction, here's a physics recap. The Ricci tensor represents an average of the Riemann curvature tensor that will encode those parts of curvature that cause volumes to shrink owing to the gravitational interaction. However, equating the Ricci tensor to the energy-momentum tensor gives a theory that doesn't conserve energy (i.e. it is incompatible with $\nabla \cdot \mathbf{T} = 0$). The Bianchi identity (geometrically) encodes the desired energy conservation. Using a trace-reversed Ricci tensor, known as the Einstein tensor, builds energy conservation back into the theory since, via the Bianchi identity, it allows the resulting equations to conserve energy.

In this example, we will temporarily reinsert the factors of c so it's a bit more clear how large various factors are.

²²Explicitly we have $R_{00} = \sum_{i=1}^3 R^i_{0i0}$.

Putting this together with $T^{00} = \rho c^2$, the Einstein equation $\mathbf{G} = \kappa \mathbf{T}$ predicts a universal field equation of

$$\vec{\nabla}^2 \Phi = \frac{\kappa c^4}{2} \rho. \quad (13.41)$$

This can be compared with Poisson's equation for gravitation (the field equation that corresponds to Newton's universal law of gravitation), which reads $\vec{\nabla}^2 \Phi = 4\pi G \rho$. Hence, we deduce that

$$\kappa = \frac{8\pi G}{c^4}. \quad (13.42)$$

In SI units, the factor is $\kappa = 8\pi G/c^4 \approx 2.08 \times 10^{-43} \text{ N}^{-1}$.

The result of the previous example, setting $c = 1$ again, is that $\kappa = 8\pi G$. We are now in a position to write a field equation for gravity, known as the Einstein field equation.²³

The Einstein field equation, version 1

In coordinates,

$$\left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) = 8\pi G T_{\mu\nu}. \quad (13.43)$$

In coordinate-free notation, this becomes

$$\mathbf{G}(x) = 8\pi G \mathbf{T}(x). \quad (13.44)$$

Example 13.5

As a further manipulation, let's take the trace of eqn 13.43. This is achieved by multiplying by $g^{\mu\nu}$, so that we get

$$g^{\mu\nu} R_{\mu\nu} - \frac{1}{2} g^{\mu\nu} g_{\mu\nu} R = 8\pi G g^{\mu\nu} T_{\mu\nu}. \quad (13.45)$$

Let's now consider the trace T of the energy-momentum tensor $T = g^{\mu\nu} T_{\mu\nu}$, and hence eqn 13.45 gives (using $g^{\mu\nu} g_{\mu\nu} = 4$ and $R = g_{\mu\nu} R^{\mu\nu}$) $R - 2R = -R = 8\pi G T$. This implies we can also write

$$R_{\mu\nu} + \frac{1}{2} g_{\mu\nu} (8\pi G T) = 8\pi G T_{\mu\nu}. \quad (13.46)$$

The previous example therefore leads to a second form for the Einstein field equation:

The Einstein field equation, version 2

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right). \quad (13.47)$$

Our method of arriving at the Einstein equation has been rather haphazard. We might ask if we can add other terms to the right-hand side of the Einstein equation. In fact, it is permissible to add any scalar multiple of the metric field tensor, so we can take our (version 1) equation $\mathbf{G}(x) = 8\pi G \mathbf{T}(x)$ and put an extra $-\Lambda \mathbf{g}(x)$ on the right-hand side and if Λ is just a constant then our new equation will still work. In fact, the property that $\nabla_\alpha \mathbf{g} = g_{\mu\nu;\alpha} = 0$ means that the divergence arguments

²³This is the equation Einstein presented in November 1915, after having rejected his first attempt.

we have been using won't be troubled by this extra term. We therefore write our third version of the Einstein equation as follows:

The Einstein field equation, version 3

$$\left(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\right) = 8\pi GT_{\mu\nu} - \Lambda g_{\mu\nu}, \quad (13.48)$$

$$\mathbf{G}(x) = 8\pi G\mathbf{T}(x) - \Lambda\mathbf{g}(x). \quad (13.49)$$

The constant of proportionality Λ is known as the **cosmological constant**. Physically, the cosmological constant represents an extra source of energy in the Universe, in addition to the matter fields. Owing to the difference in sign compared to positive $T_{\mu\nu}$ (as demanded by the dominant energy condition), a positive value of Λ will give a repulsive gravitational interaction.

In order to retain the simplicity of the equation $\mathbf{G} = 8\pi G\mathbf{T}$, we could treat the cosmological constant term as if it were a source of energy-momentum and write

$$\mathbf{G} = 8\pi G(\mathbf{T}^{\text{mat}} + \mathbf{T}^{\text{vac}}), \quad (13.50)$$

where $\mathbf{T}^{\text{mat}}$ is the energy-momentum tensor for matter that we have described in this chapter and

$$\mathbf{T}^{\text{vac}} = -\frac{\Lambda}{8\pi G}\mathbf{g} \quad (13.51)$$

is a contribution to the energy-momentum tensor from sources other than the matter fields. The introduction and rejection of Λ by Einstein has become a famous story in physics folklore. We shall discuss its consequences in Part III of the book.²⁴

We have arrived at an equation of motion for the metric field that determines gravitation. Our task is now to explore its consequences on our Universe. In the coming chapters, it is useful to keep in mind a few helpful heuristics we have seen already in dealing with general relativity.

²⁴A brief history of solutions involving Λ is given in Chapter 18.

- A metric is visualized via its light cones. These are found using $ds^2 = 0$ along null geodesics.
- We use two types of reference frame: (i) a coordinate frame, where our geometric arguments are often simplest; (ii) an orthonormal frame, where observers make measurements.
- Coordinates have no intrinsic metric significance but, in cases of spherical symmetry, useful coordinates to have in mind are spherical polars (t, r, θ, ϕ) .
- Indices are raised/lowered in the coordinate frame using the metric. Indices in orthonormal frames are manipulated with the Minkowski tensor. The connection coefficients don't vanish in the orthonormal frame. If we want this we need to shift to a local inertial frame.

- General covariance says that a valid tensor equation in flat space is a valid tensor equation in curved space. To upgrade tensor equations, exchange $\boldsymbol{\eta} \rightarrow \boldsymbol{g}$. To upgrade derivatives, use the *comma goes to semicolon* rule.
- When manipulating vectors, it's often helpful to use (i) $g_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu$; (ii) $g_{\mu\nu;\alpha} = 0$. For timelike velocities $\mathbf{u} \cdot \mathbf{u} = -1$.
- The perfect fluid, in the orthonormal frame for spherical polars, has components

$$T_{\hat{t}\hat{t}} = \rho, \quad T_{\hat{r}\hat{r}} = p, \quad T_{\hat{\theta}\hat{\theta}} = p, \quad T_{\hat{\phi}\hat{\phi}} = p.$$

Chapter summary

- The left-hand side of the Einstein field equation encodes geometry using the Ricci tensor and scalar to form the Einstein tensor $\mathbf{G}$ with components

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R. \quad (13.52)$$

- The right-hand side of Einstein's equation encodes the energy-momentum of matter fields using the tensor $\mathbf{T}$.
- These are tied together in Einstein's equation, subject to the key constraint that energy-momentum is locally conserved. The Einstein equation is $\mathbf{G}(x) = 8\pi G\mathbf{T}(x) - \Lambda\mathbf{g}(x)$ and has components

$$\left(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \right) = 8\pi GT_{\mu\nu} - \Lambda g_{\mu\nu}. \quad (13.53)$$

Exercises

- (13.1) Show that Newton's second law and his law of gravitation are invariant with respect to Galilean transformations.

- (13.2) Consider the equation

$$\vec{\nabla}^2 \left(\frac{1}{r} \right) = -4\pi\delta(\vec{x}), \quad (13.54)$$

where $r = |\vec{x}|$. We shall prove this. (a) Evaluate $\frac{\partial}{\partial x^i}r$ and $\frac{\partial}{\partial x^i}r^{-1}$. (b) Show that

$$\frac{\partial^2}{\partial x^i \partial x^j} \left(\frac{1}{r} \right) = \frac{(3n^i n^j - \delta^{ij})}{r^3}, \quad (13.55)$$

where $n^i = x^i/r$. (c) Use the previous result to show that $\vec{\nabla}^2(r^{-1}) = 0$ for $|\vec{x}| \neq 0$. (d) Defining $\vec{j} = \vec{\nabla}(r^{-1})$, use Gauss' theorem to show that

$$\int_V d^3x \vec{\nabla} \cdot \vec{j} = -4\pi, \quad (13.56)$$

where V is a volume that encloses $\vec{x} = 0$. (e) Use the previous result to complete the proof.

- (13.3) Verify that taking the trace $\text{Tr}(G^{\mu\nu})$ of the Einstein tensor with components $G^{\mu\nu}$ gives $\text{Tr}(G^{\mu\nu}) = -R$, where R is the trace of the Ricci tensor.